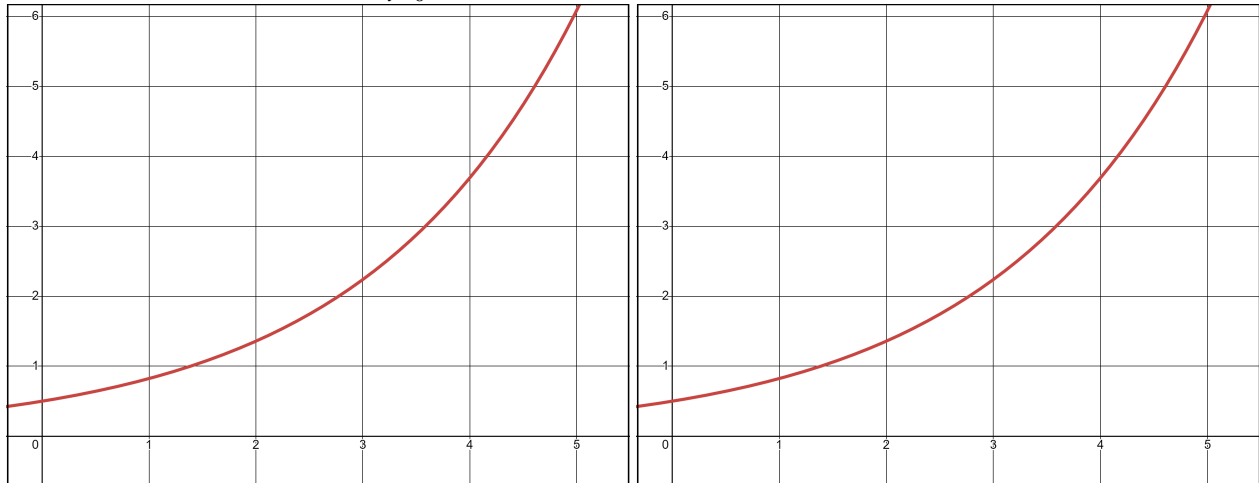


Learning target IN2 and INx, version 1

The rate at which pollution escapes a scrubbing process at a manufacturing plant increases over time as filters and other technologies become less effective. Suppose that the rate of pollution (measured in tons per week) is given by the function r that is pictured below. Throughout this problem, we're interested in $\int_{t=0}^5 r(t) dt$.



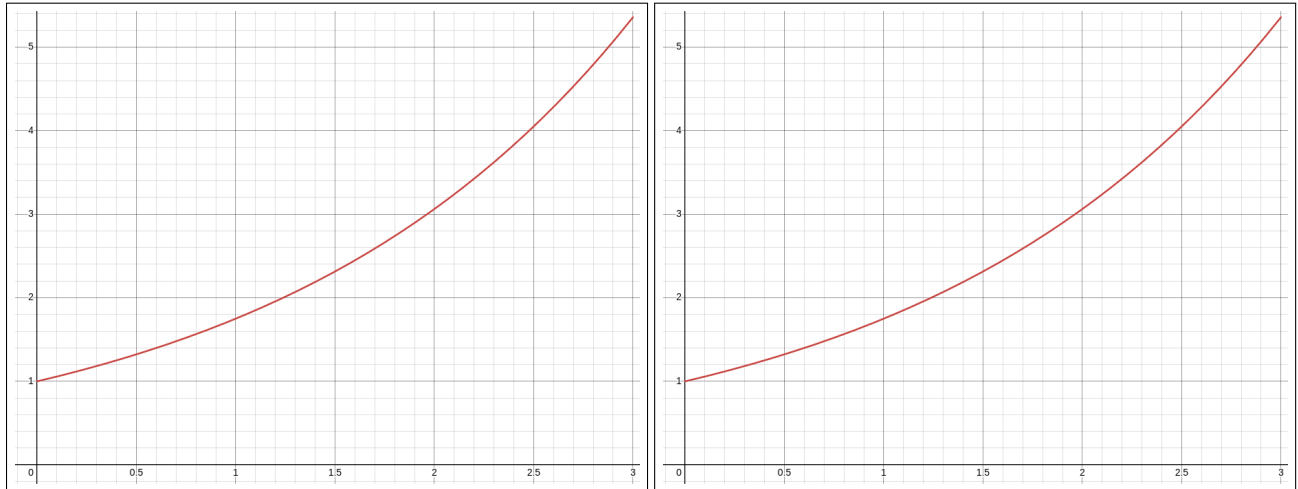
- On the first graph above, carefully draw the *left* Riemann sum with four rectangles of uniform width. (How wide should each rectangle be?)
- On the second graph above, carefully draw the *right* Riemann sum with four rectangles.
- If $r(t) = 0.5e^{0.5t}$, compute both of these Riemann sums. Show your work for computing the heights of the rectangles. (Hint: lots of your work will overlap!)

Include units on every number you write.

- Give your best guess for $\int_{t=0}^5 r(t) dt$, **include units**, and explain what this number means about pollution.

Learning target IN2 and INx, version 2

A colony of bacteria in a petri dish is growing at a rate of $f(t) = 1.75^t$ million bacteria per hour. Let's figure out how much the colony grows over the course of three hours.



- On the first graph above, carefully draw the *left* Riemann sum with four rectangles of uniform width. (How wide should each rectangle be?)
- On the second graph above, carefully draw the *right* Riemann sum with four rectangles.
- Compute both of these Riemann sums. Show your work for computing the heights of the rectangles. (Hint: lots of your work will overlap!)

Include units on every number you write.

- Give your best guess for $\int_{t=0}^3 f(t) dt$, **include units**, and explain what this number means about the bacteria colony.